

1.what is big data and what is pyspark ?

Big Data refers to very large and complex datasets that cannot be processed efficiently using traditional systems.

Characteristics (5 V's)

- **Volume** – Huge amount of data
- **Velocity** – Fast data generation
- **Variety** – Different data types
- **Veracity** – Uncertain/noisy data
- **Value** – Useful insights from data

Examples:

- Social media data
- Banking transactions
- IoT sensor data

Apache Spark **PySpark** is the Python API for Apache Spark.

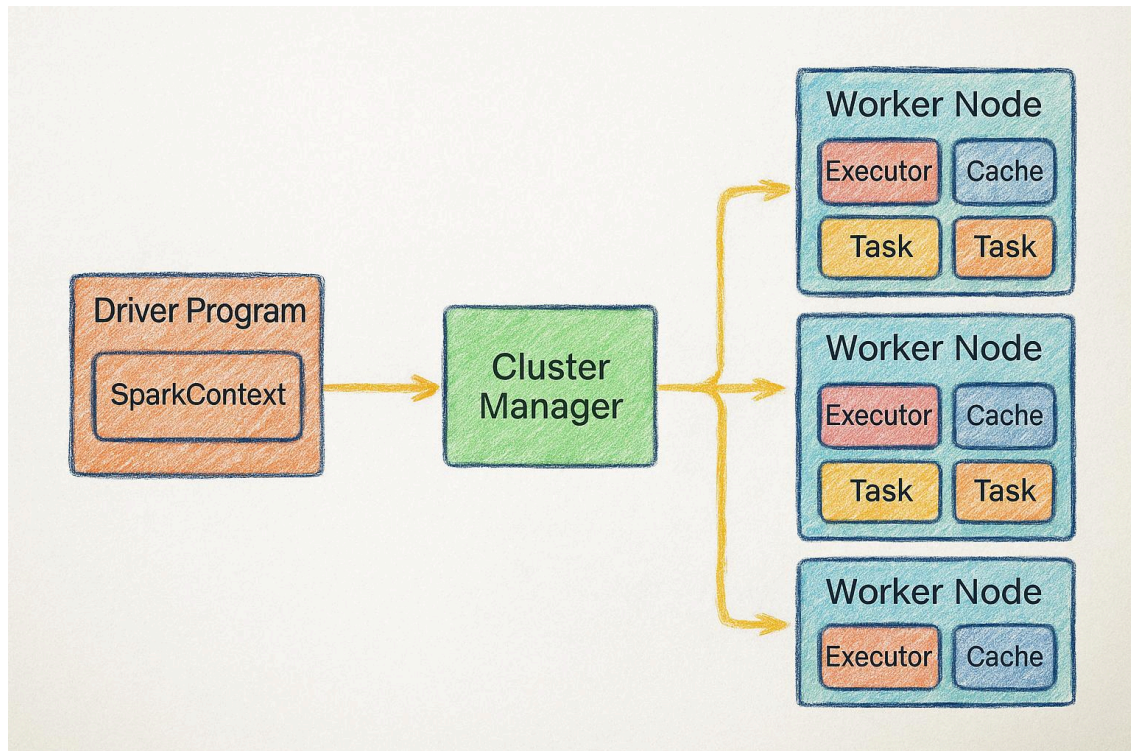
It is used for:

- Big data processing
- Distributed computing
- Data analysis
- Machine learning

Features

- Fast in-memory processing
- Parallel execution
- Scalable
- Fault tolerant

2. Spark's basic architecture



Apache Spark uses a **master-worker architecture** for distributed data processing.

Main Components

Component	Function
Driver Program	Controls the application and schedules tasks
Cluster Manager	Allocates resources to the application
Worker Nodes	Machines that perform processing
Executors	Run tasks on worker nodes

Working

1. User submits Spark application.
2. Driver program creates tasks.
3. Cluster manager allocates resources.
4. Executors process data in parallel on worker nodes.
5. Results are returned to the driver.

Features

- Fast in-memory processing
- Parallel computation
- Fault tolerance
- Scalable for big data applications

3.cluster manager, spark session, spark context, driver node, worker node

Components in Spark Architecture

Component	Description
Cluster Manager	Manages resources and allocates CPU/memory to Spark applications. Examples: Apache Hadoop YARN, Kubernetes
SparkSession	Main entry point in PySpark used for DataFrames, SQL, and Spark operations.
SparkContext	Core connection between application and Spark cluster; manages execution.
Driver Node	Main machine that runs the Spark application and schedules tasks.
Worker Node	Machines that execute tasks and process data in parallel.

4.what is client mode and cluster mode

Client Mode vs Cluster Mode in Spark

Feature	Client Mode	Cluster Mode
Driver Location	Runs on client/local machine	Runs inside cluster
Dependency on Client	Client must stay active	Client can disconnect
Performance	Slightly slower for large jobs	Better for production
Usage	Testing and development	Production environments

Client Mode

- Driver program runs on the user's machine.
- Executors run on cluster nodes.
- If client machine stops, application stops.

Cluster Mode

- Driver runs inside the cluster.
- Cluster manages the entire application.
- More reliable and scalable.